46. Active Learning

Tell me and I forget, involve me and I learn.

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Active Learning is one of those terms that sounds... almost too familiar.

"Active Learning?" → You've heard it before. Maybe in school. Maybe in life. It simply means: *learning by doing, by engaging, by being involved.*

But in Machine Learning? It takes on a much more interesting meaning.

Think about how most Machine Learning models learn: We give them data. Labeled data. Lots of it. (Images with labels; Texts with categories; Numbers with outcomes).

And the model just, learns from what we give it. Quietly. Passively. It doesn't ask questions. It doesn't challenge the data. It doesn't choose what it wants to learn from. It just accepts everything.

But here's the problem. Labeling data is expensive. It takes time, effort, human expertise. Especially in fields like: Medical diagnosis; Legal documents; Scientific research. And often, we have tons of *unlabeled* data, but only a small amount of labeled data. So now the question becomes: **What if the model could decide what it wants to learn from?**

This is where Active Learning comes in. Instead of passively receiving data, the model becomes *interactive*. It looks at a pool of unlabeled data. And starts asking:

> *"Which of these examples would help me the most... if I knew the answer?"*

Not all data is equally useful. Some examples are easy. Some are redundant. Some don't teach anything new. But some are *critical*. The model learns to identify those. And then, it asks for labels only for the most informative data points.

So instead of labeling everything. We label *strategically*. Less data. Better learning.

This changes the entire learning process. From: *"Give me everything... I'll figure it out."* → To: *"Give me the right examples... and I'll learn faster."*

If Transfer Learning was about reusing knowledge. Active Learning is about *choosing experience*. And that makes Machine Learning feel a little more human. Because learning is no longer passive. It becomes a conversation between the model and the data. And sometimes, even you.

In the next section, we'll explore how this interaction actually works.

1. Active Learning – The Overview

Now let's turn that idea of *interactive learning* into something more concrete. What exactly is Active Learning? At its core, Active Learning is about learning **efficiently**. Not by using more data. But by using *better* data.

In traditional supervised learning, the process is very straightforward. We prepare a dataset. We label everything. Then we train the model. Simple. But also very expensive.

Because labeling data is not free. It takes time. It takes effort. And sometimes, it requires experts. And here's the uncomfortable truth: Most of that labeled data?

→ Not all of it is useful. Some examples are repetitive. Some are obvious. Some don't really help the model learn anything new.

So now we ask a better question: **What if we didn't label everything?** What if the model could *choose* what it wants to learn from?

That's exactly what Active Learning does. Instead of passively learning from a fixed dataset… the model becomes selective. It looks at a large pool of **unlabeled data**… And starts asking:

> "Which of these examples would actually help me improve?"

Then, it requests labels only for those specific data points. Not all data. Just the *most informative* ones.

So the process becomes iterative:

- The model learns from a small labeled set. It evaluates unlabeled data. It selects the most useful samples. It asks for their labels. And then learns again

Over and over. Each step makes the model smarter. Without needing to label everything.

And here's the key idea behind it all: ***"If a model can choose what it learns from… it can learn faster with less data."***

But how does the model decide what is "useful"? This is where things get interesting. Active Learning focuses on **uncertainty**. The model pays attention to cases where it is unsure. Where predictions are not confident. Where the boundary between decisions is unclear. Because those are the moments where learning happens. Not when things are obvious, but when things are confusing.

So instead of reinforcing what it already knows, the model actively targets what it *doesn't* understand. And that's what makes it powerful.

Let's make this more intuitive:

Imagine a survival game. You are surrounded by fruits. Some are safe. Some are poisonous. And you have a problem: You don't know which is which. The only way to find out… is to taste them. But if you choose wrong, you die.

Now suppose you are given one rule: ***"All red fruits are poisonous."*** *→ That changes everything. You won't randomly pick fruits anymore. You'll avoid red ones, and carefully choose others that help you learn more about the patterns.*

You'll ask smarter questions like: Does size matter? Does shape matter? What about color combinations? You don't test everything. You test what gives you the most information.

That's Active Learning. Not random exploration. But **strategic learning**. So in the end, Active Learning is not about having more data. It's about asking better questions. Choosing the right examples. Focusing on uncertainty. And learning where it matters most.

Because sometimes, learning less, Is actually how you learn more.

3. Active Learning – Workflow and Core Concepts

Now that we understand the idea behind Active Learning... let's see how it actually works in practice. Because this is not just a concept. It's a process. A loop. A conversation between the model and the data.

Active Learning Workflow

At a high level, Active Learning follows a cycle. Not a straight line. Not a one-time training step. But an **iterative process** that keeps improving over time.

- We start simple. First, we take a **small labeled dataset**. Not everything. Just enough to get the model started. We train an initial model using this limited data.
    - At this point, the model knows something. But not much.
- Next, we expose it to the real world. A large pool of **unlabeled data**. And the model does what it always does: It makes predictions.
    - But this time, something is different. We don't fully trust those predictions. Instead, we ask: **"How confident are you?"**
    - This is where uncertainty comes in. For every prediction, the model gives us a sense of confidence. Sometimes it's very sure. Sometimes, not at all. And that uncertainty becomes our guide.
- Now comes the key step. We don't pick data randomly. We select the samples where the model is most uncertain, most confused, closest to making a mistake. These are the most valuable learning opportunities.
- Once selected, we ask for help. This is where the **oracle** comes in. The oracle could be a human annotator, a domain expert, or a trusted system.

    - Its job is simple: Provide the **correct label**.
- Now we take those newly labeled examples, Add them back into the training data. And retrain the model. And just like that, the model improves.
- Then? We repeat. Again, and again, and again. Until we reach a stopping point:
    - We run out of labeling budget
    - The model is good enough or improvements stop increasing

So the workflow becomes:

> ***Train*** → ***Predict*** → ***Measure uncertainty*** → ***Select*** → ***Label*** → ***Retrain*** → ***Repeat***

A loop that keeps getting smarter, without needing to label everything.

Key Concepts of Active Learning

Now let's break down the ideas that make this process work.

- **Query Strategy — "How do we choose?"**

Not all uncertain data is equally useful. So we need a strategy. A rule that decides: Which samples should we label next?

Some strategies focus on uncertainty (most confused samples), diversity (different types of data), representativeness (samples that reflect the dataset well).

This choice defines how the model learns.

- **Model Uncertainty — "Where am I unsure?"**

This is the heart of Active Learning. The model doesn't just predict. It evaluates its own confidence → ***Low confidence = high learning potential.***

These are the moments where the model says: *"I don't understand this yet."* → And those are exactly the points we target.

- **Oracle — "Who gives the truth?"**

The model doesn't label data itself. It asks. And the oracle answers. In real-world systems... this is often a human expert, a trained annotator, or a trusted labeling system

Without the oracle, the learning process cannot move forward.

- **Stopping Conditions — "When do we stop?"**

Since this is an iterative process, we need a way to stop it.

Common stopping points include:

1. Reaching a desired accuracy
2. No significant improvement over time
3. Running out of labeling resources

Because even smart learning has limits.

When you step back, Active Learning is doing something very powerful. It is not trying to learn from everything. It is trying to learn from the **right things**. By focusing only on the most informative samples. It reduces labeling cost, human effort, data requirements. While still improving performance.

And that's the real shift. Not more data. But better choices. Because in Active Learning: ***"What you choose to learn from… matters more than how much you learn"***

3. Active Learning – The Strategy with Example

Now let's bring everything together. Because until now, Active Learning sounds smart. But the real question is: ***How does it actually behave in a real situation?***

Let's go back to our survival game. We start with one rule: ***Red fruits → Poisonous***. So our initial labeled dataset looks like this:

Type	Size	Color	Poisonous
Berry	Small	Red	Yes
Apple	Medium	Red	Yes
Cherry	Small	Red	Yes

That's it. Very small. Very limited.

Now imagine doing this with **traditional learning**. You randomly pick fruits, taste them, label them. That's dangerous. And inefficient. Because:

- You might pick risky samples
- You might waste effort on uninformative data

- You label more than necessary

Active Learning does the opposite. It doesn't guess randomly. It chooses *strategically*.

Active Learning Strategy - Here's how the process unfolds.

Step 1: Initial Model

From the small dataset, the model learns a simple rule:

- Red → Poisonous
- Everything else → Unknown

At this stage, the model is weak. It knows something. But not enough.

Step 2: Unlabeled Pool

Now we introduce new fruits:

Type	Size	Color	Label
Apple	Large	Green	?
Berry	Small	Blue	?
Mango	Large	Yellow	?
Cherry	Small	Dark Red	?

The model tries to predict, even without knowing the truth.

Step 3: Prediction + Confidence

Now something important happens. The model doesn't just predict. It also says: ***"How sure am I?"***

Fruit	Prediction	Confidence
Green Apple	Safe	0.55
Blue Berry	Safe	0.51
Yellow Mango	Safe	0.95
Dark Red Cherry	Poisonous	0.99

Now look carefully. Which ones are useful? Not the confident ones. Not: Yellow Mango (0.95); Dark Red Cherry (0.99)

Those don’t teach much. The real value is here: Blue Berry (0.51); Green Apple (0.55). This is where the model is confused.

Step 4: Query the Oracle

So the model asks: “What is the label of Green Apple?” The oracle answers: ***Green Apple → Safe***

Now we update the dataset, retrain the model, and repeat.

Iterative Learning

This cycle continues: ***Train*** → ***Predict*** → ***Find uncertainty*** → ***Ask*** → ***Learn*** → ***Repeat***

And over time… the model discovers patterns like:

- Red → Poisonous
- Large Green → Safe
- Small Blue → (learned through iteration)

And here’s the magic: We didn’t label everything. We only labeled what mattered.

How Confidence is Calculated in Active Learning

Now let’s go deeper. Because the entire strategy depends on one thing: **Measuring uncertainty.**

1. Uncertainty Sampling - This is the simplest approach. We look at the highest predicted probability:

$$Confidence(x) = \max_y P(y|x)$$

If this value is low → the model is uncertain.

Example:

Fruit	Safe	Poisonous	Confidence
Green Apple	0.55	0.45	**0.55**
Blue Berry	0.51	0.49	**0.51**
Yellow Mango	0.95	0.05	**0.95**

→ Blue Berry is the most uncertain.

2. Margin Sampling - Here we look at the gap between top two probabilities:

$$Margin = P_1 - P_2$$

Small margin → high uncertainty.

Example:

Fruit	Margin
Green Apple	0.10
Blue Berry	0.02
Yellow Mango	0.90

→ Blue Berry again = most uncertain.

3. Entropy Sampling

Now we measure overall uncertainty:

$$H(x) = -\sum P(y|x) \log P(y|x)$$

Higher entropy → more confusion.

Example:

Fruit	Safe	Poisonous	Entropy
Green Apple	0.55	0.45	$-[0.55 \times log(0.55) + 0.45 \times log(0.45)] = 0.992 \rightarrow High$
Blue Berry	0.51	0.49	$-[0.51 \times log(0.51) + 0.49 \times log(0.49)] = 0.999 \rightarrow VeryHigh$

Yellow Mango	0.95	0.05	$-[0.95 \times log(0.95) + 0.05 \times log(0.05)] = 0.286 \rightarrow Low$

4. Query by Committee

Now we go one step further. Instead of one model, we use multiple models. Each one votes.

Fruit	Model A	Model B	Model C	Result
Green Apple	Safe	Safe	Poisonous	Disagreement
Yellow Mango	Safe	Safe	Safe	Agreement

Any disagreement = uncertainty. → So we label **Green Apple**.

All these strategies are different ways of answering one question: **"Where is the model most confused?"**

Because that's where learning happens. Active Learning is not about more data. It's about **better questions**. And once the model learns to ask the right questions, it doesn't just learn faster. It learns smarter.

4. Active Learning – The Applications

Active Learning stands out from traditional workflow by focuses on informative samples instead of random samples. This means instead of treating every unlabeled data point as equally important, the model actively asks a more strategic question: *"Which examples will teach me the most if I learn them next?"*

In a normal supervised learning setup, we assume a fixed labeled dataset and train the model on whatever we already have. But in many real-world problems, labeling data is expensive, slow, or requires expert knowledge. Active Learning changes this dynamic. The model is no longer passive—it becomes part of the data collection process. It evaluates the unlabeled pool, identifies the most uncertain or confusing examples, and requests labels only for those. In a sense, the model is not just learning from data—it is *choosing what to learn from*.

This creates a feedback loop that is very similar to human learning. When you study, you don't randomly revise everything equally. You naturally focus more on questions you get wrong or concepts you are unsure about. Active Learning formalizes this idea inside machine learning systems.

The strategy is typically driven by uncertainty. The model looks for data points where its predictions are weakest or most unstable. These “high-value” samples are then sent for labeling and added back into the training set. Over multiple iterations, the model gradually builds a stronger understanding while using significantly fewer labeled examples.

From a practical standpoint, this makes Active Learning extremely powerful in scenarios where data is abundant but labels are not. Instead of spending resources labeling everything, we prioritize only the most informative cases. The result is a system that learns faster, costs less to train, and often achieves strong performance even when labeled data is limited.

Some common application areas include:

- Medical diagnosis, where only the most uncertain scans (like X-rays or MRI images) are sent to experts for annotation
- Fraud detection, where suspicious transactions are prioritized for manual investigation
- Image annotation, where labeling effort is focused on images that improve model understanding the most
- Scientific research, where experiments are selected based on which ones are expected to provide the highest information gain

In essence, Active Learning shifts the learning process from *“learn everything given to you”* to *“learn what matters most”*. It turns data collection into an intelligent decision-making problem, making machine learning systems more efficient and closer to how humans naturally learn under constraints.

5. Active Learning with Python

Active Learning stands out from traditional workflow by focuses on informative samples instead of random samples. This means the model does not treat every unlabeled data point equally. Instead, it actively searches for the most uncertain or informative examples and asks for their labels first. In this way, the model learns in a more targeted and efficient manner, similar to how humans focus on concepts they are unsure about rather than revising everything blindly.

To demonstrate this idea, we use a simulated dataset:

[fruit_active_learning_dataset.csv]

The dataset contains around 1020 samples, where each fruit is described using three categorical features:

- **type** → type of fruit
- **size** → small, medium, large
- **color** → red, green, yellow, blue

The target label is:

- **poisonous** → whether the fruit is poisonous or not

In this simplified rule-based setup, all **red fruits are considered poisonous**, while other colors may be safe or uncertain. This controlled structure allows us to clearly observe how Active Learning behaves.

Step 1: Load the Dataset

```
import pandas as pd

# Load dataset
df = pd.read_csv("fruit_active_learning_dataset.csv")
```

Step 2: Convert Categorical Data (One-Hot Encoding)

Machine learning models cannot directly understand categorical values, so we convert them into numerical form using One-Hot Encoding.

```
# One-hot encoding for categorical features
encoded_df = pd.get_dummies(df[["type", "size", "color"]]).
replace({True: 1, False: 0})

# Add target column
encoded_df["poisonous"] = df["poisonous"]
```

Step 3: Prepare Features and Target

We separate input features and labels, then convert them into NumPy arrays for efficient computation.

```
target = "poisonous"
features = encoded_df.columns.drop(target)

# Convert to NumPy arrays
X_encoded = encoded_df[features].values
y = encoded_df[target].values

print("Encoded Features Shape:", X_encoded.shape)
print("Target Shape:", y.shape)
```

Encoded Features Shape: (1020, 11)
Target Shape: (1020,)

Step 4: Split into Labeled and Unlabeled Sets

Unlike traditional machine learning, Active Learning starts with a small labeled dataset and a large pool of unlabeled data.

```
from sklearn.model_selection import train_test_split

# 25% labeled, 75% unlabeled pool
labeled_set, unlabeled_set = train_test_split(X_encoded, te
st_size=0.75, random_state=42)

print("Labeled Set Length:", len(labeled_set))
print("Unlabeled Set Length:", len(unlabeled_set))
```

Labeled Set Length: 255
Unlabeled Set Length: 765

We then separate features and labels:

```
# Split features and labels
X_labeled = labeled_set[:, :-1]
y_labeled = labeled_set[:, -1]

X_unlabeled = unlabeled_set[:, :-1]
y_unlabeled = unlabeled_set[:, -1]  # only for evaluation
```

Step 5: Train a Model (Logistic Regression)

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report

model = LogisticRegression()
```

Step 6: Uncertainty Sampling Strategy

Active Learning relies on selecting the most uncertain samples. We measure uncertainty using prediction probabilities.

```
import numpy as np

def uncertainty_sampling(model, X_pool, k=5):
    probs = model.predict_proba(X_pool)

    # uncertainty = 1 - max confidence
    uncertainty = 1 - np.max(probs, axis=1)
    k = min(k, len(X_pool))

    # select top-k most uncertain samples
    query_idx = np.argsort(uncertainty)[-k:]

    return query_idx
```

Step 7: Active Learning Loop

This is where the system actively learns by iteratively selecting new samples.

```
iterations = 10

for i in range(iterations):
    print(f"\nIteration {i+1}")

    # Train model on labeled data
    model.fit(X_labeled, y_labeled)

    # Select uncertain samples
    query_idx = uncertainty_sampling(model, X_unlabeled, k=
5)

    # Add selected samples to labeled set
    X_labeled = np.vstack((X_labeled, X_unlabeled[query_id
x]))
    y_labeled = np.concatenate((y_labeled, y_unlabeled[quer
y_idx]))

    # Remove them from unlabeled pool
    X_unlabeled = np.delete(X_unlabeled, query_idx, axis=0)
    y_unlabeled = np.delete(y_unlabeled, query_idx, axis=0)

    print("Added samples:", len(query_idx))
    print("Labeled set size:", len(X_labeled))
    print("Remaining unlabeled:", len(X_unlabeled))
```

Run the code to see the output

Step 8: Final Evaluation

After multiple iterations, we evaluate the model on the remaining unlabeled data.

```
y_pred = model.predict(X_unlabeled)
```

```
print("\nClassification Report on Remaining Unlabeled Se
t:")
print(classification_report(y_unlabeled, y_pred))
```

What Happens During Learning?

In each iteration:

- The model is trained on a small labeled set
- It predicts probabilities on unlabeled data
- It identifies the most uncertain samples
- Those samples are sent for "labeling"
- They are added back into training data

This simulates a real-world scenario where an oracle (human expert or system) labels only the most informative cases.

Over time, the model does not learn randomly—it learns strategically.

Summary

Active Learning is, at its core, learning with intention. Instead of treating all data equally, the model constantly asks: *"Which examples do I not understand yet?"* and focuses only on those. This turns the training process into a feedback loop between uncertainty and improvement.

Rather than consuming large labeled datasets, the model gradually builds knowledge from a small but carefully selected set of examples. Each iteration refines its understanding, making the learning process more efficient, more targeted, and often more accurate.

In simple terms: Active Learning is not about learning more data. It is about learning the right data at the right time.

References

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Next Section:

🤖 47. T-Distributed Stochastic Neighbor Embedding